

COURSE CODE/TITLE: CPE 010-CPE21S1 - Data Structures and Algorithms

SECTION: CPE21S1

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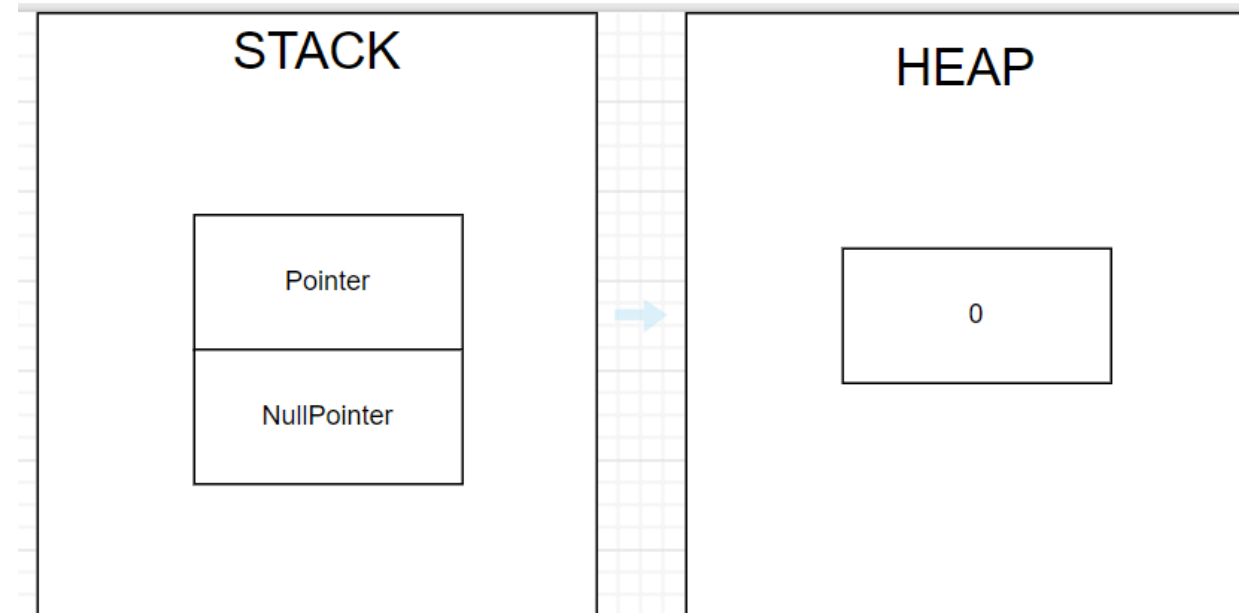
INSTRUCTOR: Engr. Jimlord Quejado

ACTIVITY NO. 2_ Seatwork 2.1: Memory Diagram Exercise

1. PROCEDURE OUTPUT

Draw the memory diagram for the following:

```
1. int *ptr = nullptr;  
>
```



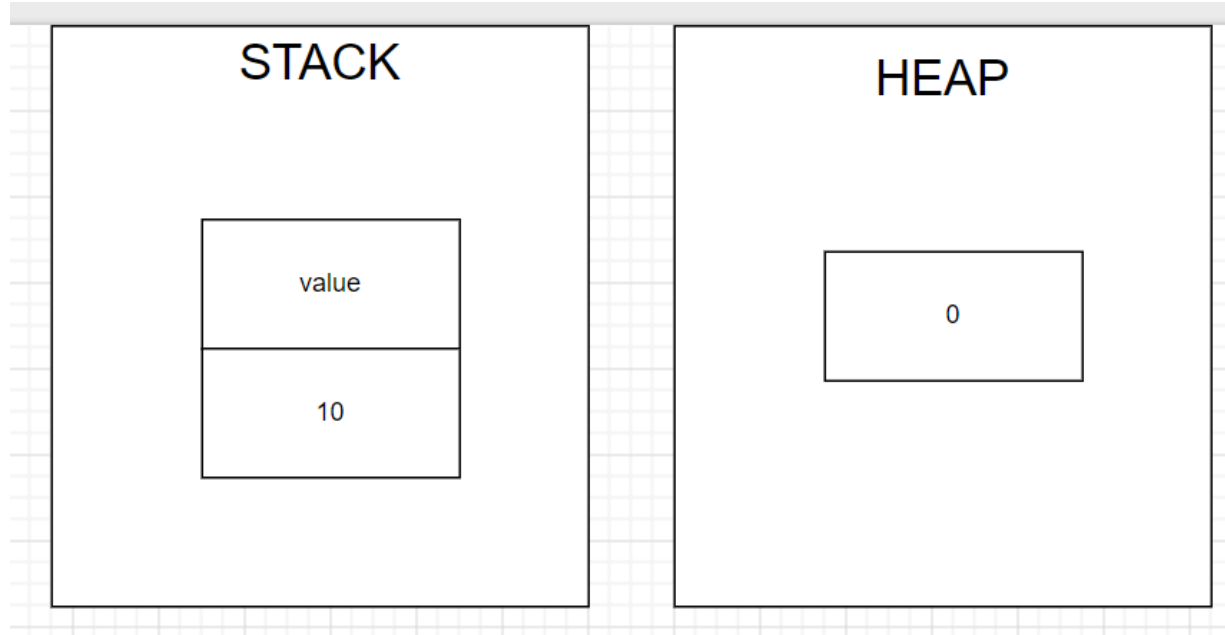
DESCRIPTION:

>> A pointer Variable is created in the stack memory

>> It is then initialized with null pointer thus it does not lead to any location

```
2. int value = 10;  
   int *ptr= &value;  
>
```

1. PROCEDURE OUTPUT

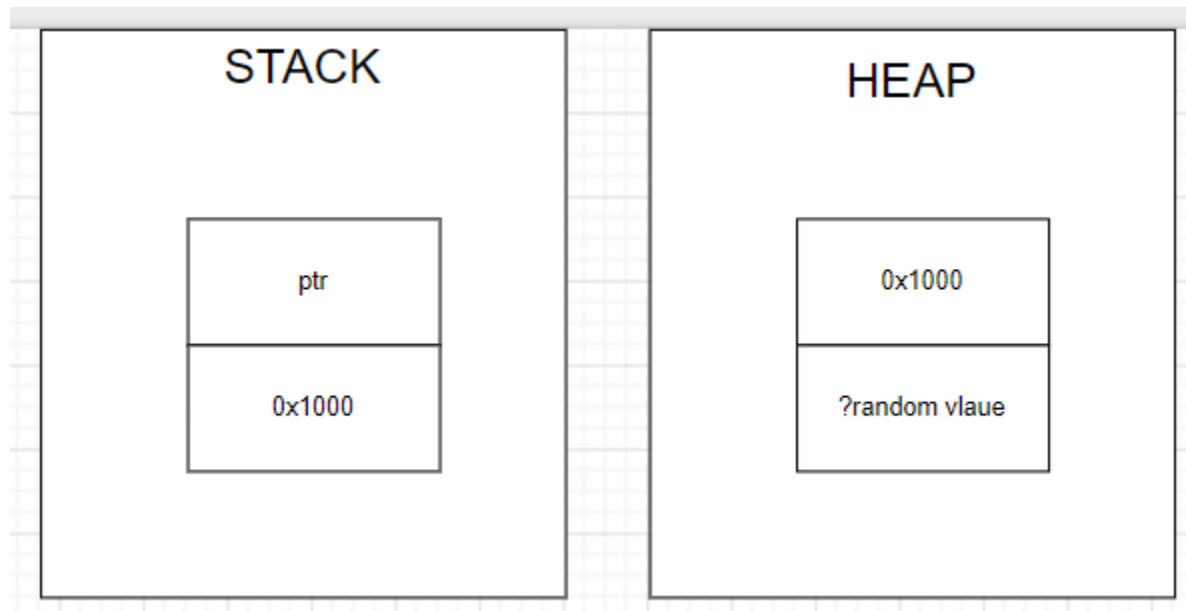


>> An int variable was created and has a value of 10

>> the heap is still empty because there is no new variable for it.

```
3. int *ptr = new int;
```

>

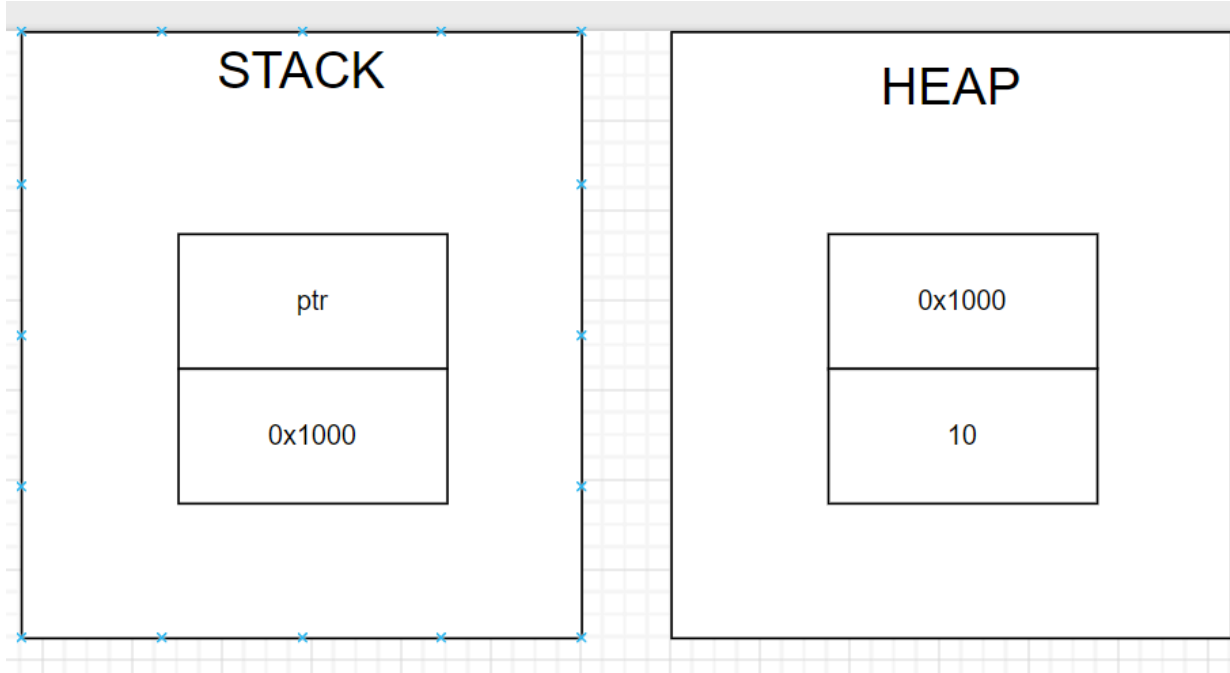


>> A pointer was created and stored in the stack, and since we always start(as far as the knowledge knows), at 0x1000, the address is located there.

>> And since there are no assigned value, the integer always produce a random value from positive to negative.(heap)

ACTIVITY NO. 2_**Seatwork 2.1: Memory Diagram Exercise****1. PROCEDURE OUTPUT**

```
4. int *ptr = new int(10);  
>
```



>> Like the number 3, it is basically the same, that the address starts at 0x1000, however since we have the value/integer of 10, the value in the heap is also 10.

5. Differentiate Constructor, Destructor, Copy Constructor, Copy Assignment Constructor.

> Constructor - Initializes an object once it is created. It also allocates resources for the object.

> Destructor - the purpose of destructor was to free up the resources used in the code when an object is destroyed

> Copy Constructor - Creates a new object that copies the contents/values of the pre existing and targeted object to the newly created one. Basically a replica of an old painting and being drawn today.

> Copy Assignment Operator - basically copies the content of an old object and places/pastes them to a same old but different object. Basically 2 folders are created in 2020, then at folder 1 you copied all the contents, then places them at folder 2, even if folder 2 has its pre existing contents.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY**3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT**

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: ChatGPT

1. PROCEDURE OUTPUT

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: To have a more in depth explanation and explanation about what to do/ what is the item being searched.

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: It did not replace my thought for basically it just tell me what is being searched or what to do within the activity, and every explanation was mine and mine alone from the referred information that I learned on both the discussion of our professor and chatgpt's clarification.

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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